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Highly Water-Resistant La-Doped Co_3O_4 Catalyst for CO Oxidation

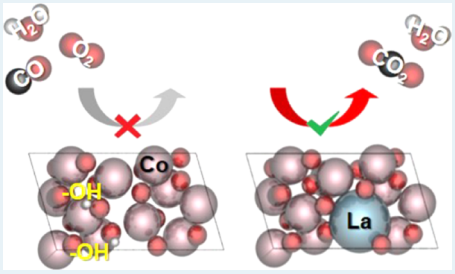
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Supporting Information

ABSTRACT: Co_3O_4 is an attractive alternative to precious metal catalyst for CO oxidation due to its low cost and earth abundance, but its high catalytic activity is severely degraded in the presence of water. Here, we show that doping La into Co_3O_4 surfaces significantly enhances CO oxidation activity under moisture-rich condition. While bare Co_3O_4 is deactivated under moisture, the La-doped Co_3O_4 catalysts exhibit greatly improved activity and water resistance. The higher ratio of active Co^{3+} on the La-doped Co_3O_4 surface results in the enhanced activity. Especially, the La doping greatly reduces the formation of surface OH on the Co_3O_4 surface, which causes the poor water resistance. Density functional theory calculations reveal that the La doping suppresses the vacancy-assisted dissociative adsorption of H_2O, resulting in less formation of surface OH and thereby mitigating water poisoning on the Co_3O_4 surface. This work can provide an insight into the surface restructuring for highly active and water-resistant Co_3O_4 catalysts.



KEYWORDS: *Co_3O_4, CO oxidation, La doping, water resistance, surface OH*

1. INTRODUCTION

Platinum group metals (PGMs) are widely used in heterogeneous catalysis due to their high activity, but the industrial application of PGMs in a large scale has been limited by their high cost and low supply.[1,2] Alternatively, transition-metal oxides have been of interest as a low-cost catalyst with good low-temperature activity for CO oxidation.[3] In particular, Co_3O_4 is considered to be the most active material due to its excellent redox properties.[3−5] By controlling the facets and morphologies of Co_3O_4 nanostructure, certain coordination between Co cations and oxygen anions can be deliberately exposed on the surface for achieving high activity.[6−11] Octahedrally coordinated Co^{3+} is known as an active site, while tetrahedrally coordinated Co^{2+} is assumed to be almost inactive.[12,13] It has been reported that Co_3O_4 nanorods with preferentially exposed (110) planes exhibit an excellent catalytic activity for cryogenic CO oxidation due to the high concentration of Co^{3+} on the surface.[12] Additionally, substituting inactive Co^{2+} in Co_3O_4 nanowires with other components could enhance the CO oxidation activity.[14] Increasing the amount of Co^{3+} exposed on the surface can be crucial for enhancing low-temperature activity.[15−17]

Although Co_3O_4 catalyst is highly active for CO oxidation, the presence of water vapor in the reactant gas severely degrades its activity.[4,12,18,19] Various deactivation mechanisms by water on Co_3O_4 have been proposed; the adsorption of reactants can be inhibited by H_2O molecules adsorbed on Co^{3+} active sites[19,20] or hydroxyl groups formed by dissociation of H_2O on the surface.[12,18] Additionally, H_2O can promote the formation of carbonate species, which poison the catalyst surface.[21] Recently, a systematic theoretical study has been reported on all of the possible deactivation pathways by water on Co_3O_4.[22] It was found that gaseous O_2 adsorption can be suppressed by the surface lattice OH generated by dissociative adsorption of H_2O on oxygen vacancy, and the activity of the surface lattice OH is less than that of surface lattice oxygen. Moreover, the stable bicarbonates formed on the surface can block the Co^{3+} active sites, resulting in additional deactivation.[22] Therefore, it is highly important to enhance water resistance of Co_3O_4 for stable catalytic performance.

In an effort to improve water resistance of Co_3O_4, a few composite oxides have been developed. For example, the formation of composite materials with another metal oxide such as CeO_2 and SnO_2 can reduce water poisoning because of strong interaction between Co_3O_4 and the other metal oxides.[23−25] In another strategy, water adsorption can be effectively suppressed by producing hydrophobic catalyst surface, which can be achieved by combining with hydrophobic materials such as poly(dimethylsiloxane)[26] and carbon nanotubes.[27] A recent study also suggests that rare-earth oxides (REOs) are inherently hydrophobic owing to their surface electronic structure.[28] REOs have a $5s^2p^6$ outer shell with full octet of electrons protecting the unfilled 4f orbitals of the inner shell, thus they hardly form hydrogen bonds with water molecules on the surface. However, surface stoichiometry, surface relaxations, the adsorption of contaminants from the

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atmosphere, and the formation of surface hydroxyl groups would affect the hydrophobic property of the REO surface.[29−32]

In this work, we demonstrate greatly enhanced CO oxidation activity under moisture-rich condition by doping La into the Co_3O_4 phase. The enhanced activity of the La-doped Co_3O_4 was attributed to the increase of active Co^{3+} on the surface. Especially, the La doping significantly reduced the formation of surface OH on the Co_3O_4 with improved water resistance. The effect of La doping on the catalytic property of Co_3O_4 was explored in detail, and the suppressed formation of surface OH on the La-doped Co_3O_4 was elucidated using density functional theory calculations.

2. EXPERIMENTAL SECTION

2.1. Catalyst Syntheses. Bulk Co_3O_4 nanoparticles were synthesized by coprecipitation. Cobalt nitrate hexahydrate ($Co(NO_3)_2{\cdot}6H_2O$, 0.25 mmol, Sigma-Aldrich) was added in 25 mL of deionized (DI) water. Under vigorous stirring, 25 mL of sodium hydroxide solution (NaOH, 0.375 M) was instilled dropwise into the initial precursor solution and then the solution was stirred for 2 h. The precipitate was filtered with DI water and ethyl alcohol. After drying at 50 °C for 12 h in a vacuum oven, the resulting cakes were crushed into fine powder in a mortar. The powders were calcined in static air at 500 °C for 5 h. La-doped Co_3O_4 catalysts were synthesized using the same procedure except that lanthanum nitrate hexahydrate ($La(NO_3)_3{\cdot}6H_2O$, Sigma-Aldrich) was added in 25 mL of DI water with an adjusted molar concentration (La = 10, 20, and 30 mol %). $LaCoO_3$ perovskite was synthesized using the same method with a Co/La molar ratio of 1:1. La_2O_3 nanoparticles were synthesized for characterizations using the same method without Co precursor.

2.2. Catalytic Reactions. CO oxidation was performed using a U-shaped quartz fixed-bed reactor (internal diameter = 15 mm, height = 60 mm) with 50 mg of the catalyst. The reactor temperature was measured by a thermocouple located close to the catalyst layer. The sample was pretreated at 100 °C under He flow for 1 h and cooled down to room temperature. CO oxidation under dry condition was tested at a space velocity of 120 000 mL g^{-1} h^{-1} with a feed stream consisting of 0.4% CO, 10% O_2, and balance He. The feed gas was passed through a water bubbler to achieve wet condition, and the water content was varied by controlling the temperature of the bubbler. The reaction temperature increased by 10 °C at a ramp rate of 5 °C min^{-1}, and the catalyst was held at each temperature in a continuous feed flow for 10 min to reach steady state. The outlet gases were analyzed by an on-line gas chromatograph (GC, Younglin GC 6500) equipped with Carboxen1000 column (75035, SUPELCO, 15 ft × 1/8 in. × 2.1 mm) and thermal conductivity detector. Simulated exhaust oxidation test was also conducted at 120 000 mL g^{-1} h^{-1} with a feed flow consisting of 0.4% CO, 0.1% C_3H_6, 0.1% C_3H_8, 5% H_2O, 10% O_2, and balance He. The catalysts were compared to a commercial diesel oxidation catalyst (DOC), containing 1.1 wt % Pt, 0.9 wt % Pd, and 0.1 wt % Rh, provided by Hyundai Motors. All experiments were carried out at atmospheric pressure.

2.3. Characterizations. The Brunauer−Emmett−Teller surface area was investigated by measuring N_2 adsorption at −196 °C in a Tristar II 3020 (Micromeritics). The samples were degassed overnight at 100 °C under vacuum before the measurements. The crystalline structure was examined by X-ray diffraction (XRD; Rigaku, Cu Kα radiation). The scan angle extended from 20 to 80° using a step size of 0.01° at a rate of 5° min^{-1}. The powder samples were deposited on a 0.2 mm deep glass holder. X-ray photoelectron spectroscopy (XPS; K-alpha, ThermoVG Scientific) was performed to investigate the surface state at room temperature under 4.0×10^{-6} Pa using a monochromatic Al source (1486.6 eV). The maximum intensity of C 1s signal was set to be 284.8 eV as a reference. The XPS samples were prepared by uniformly mounting the catalysts on an adhesive Cu tape to form a condensed film. Transmission electron microscopy (TEM; a Tecnai G2 F30 S-Twin, FEI) was measured with a voltage of 300 kV. The high-angle annular dark-field scanning TEM (HAADF-STEM) images and the energy-dispersive X-ray spectroscopy (EDS) mapping images were also taken using a Talos F200X (FEI) with 200 kV. The TEM samples were prepared by dispersing the catalysts into ethanol, then dropping them onto a lacey-carbon film supported on Cu grid (300 mesh, Electron Microscopy Sciences). Contact angles were investigated using a contact angle meter (Phoenix 150, SEO) by dropping 10 μL of DI water on the surface of the carbon tape coated with the sample at room temperature.

Diffuse reflectance infrared Fourier transform (DRIFT) spectroscopy was conducted on a Nicolet iS-50 (Thermo Fisher Scientific). The catalyst (20 mg) was put into a sample cup and set inside a DRIFT cell fitted with a ZnSe window. To investigate the water adsorption, the catalysts were exposed in water-saturated He at room temperature for 30 min and purged with He flow for 30 min to remove physically adsorbed water on the surface prior to the IR measurements. In situ DRIFT spectroscopy for CO oxidation was performed to investigate the formation of surface OH and bicarbonate species during the reaction. The catalyst was pretreated at 100 °C under Ar flow for 1 h and cooled to room temperature. Under dry condition, the catalysts were exposed to a feed stream consisting of 1% CO, 10% O_2 balanced with Ar for 10 min at room temperature, and then the IR spectrum was obtained. Under wet condition, the feed gas was passed through a water bubbler at room temperature (~3% of water vapor), and then the IR spectrum was obtained.

2.4. Computational Calculation. Density functional theory (DFT) calculations were performed using the Vienna ab initio simulation package.[33] The exchange−correlation functional of the Kohn−Sham equation within DFT formalism was considered by the spin-polarized generalized gradient approximation developed by Perdew, Burke, and Ernzerhof.[34] Projector augmented wave method with a cutoff of 500 eV was employed to efficiently replace the core electrons, and the Brillouin zone was sampled with 3 × 5 × 1 k-points grid within the scheme of Monkhorst and Pack.[35] Dipole correction was employed to treat the physically spurious local potential induced by the use of asymmetric slab model. The localization of strongly correlated electrons and their unrealistic self-interaction in the conventional DFT calculations were accounted for by the DFT + U method.[36] The effective U value of 2 eV was added to correctly describe the d-electrons of Co ions.[37−39] The geometric structures of the surface models were optimized using the conjugate gradient method with a maximum atomic force of <0.03 eV $Å^{-1}$. Based on the results of XRD patterns, the (311) surface of Co_3O_4 was used in the DFT calculations. Figure S1 shows the optimized model, and a vacuum thickness of >16 Å was introduced to prevent the interactions between fictitious periodic slabs along [311].

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The oxygen vacancy formation energy (E_{vf}), which is the energy required for an oxygen to desorb from the Co_3O_4 surface, was calculated as

$$E_{vf} = E_{slab+V_O} + 1/2E_{O_2} - E_{slab}$$

where E_{slab+V_O}, E_{O_2}, and E_{slab}, are the total energies of the slab model with an oxygen vacancy on the surface, gas-phase O_2, and the slab without oxygen vacancy, respectively. The adsorption strength of an H_2O molecule, which adsorbs on the oxygen vacancy site in a dissociative manner to form two surface OH groups, was estimated by the adsorption energy (E_{ads}) calculated as

$$E_{ads} = E_{OH/slab} - E_{H_2O} - E_{slab+V_O}$$

where $E_{OH/slab}$ and E_{H_2O} are the total energies of the slab model with surface hydroxyl groups and gas-phase H_2O molecule, respectively.

3. RESULTS AND DISCUSSION

Figure 1 shows the catalytic activity of bare Co_3O_4, La-doped Co_3O_4 catalysts, $LaCoO_3$, and bare La_2O_3 for CO oxidation. Under dry condition, as shown in Figure 1a, bare Co_3O_4 achieved 90% CO conversion at 162 °C, which is comparable to the reported CO oxidation activity in Co_3O_4 nanoparticles.[5,6,40,41] All La-doped Co_3O_4 catalysts showed better activity than bare Co_3O_4, and 20% La-doped Co_3O_4 has the highest catalytic activity by achieving 90% CO conversion at 136 °C. $LaCoO_3$ perovskite showed slightly improved activity at low-temperature light-off compared to bare Co_3O_4, but the activity was lower in the high CO conversion region. Bare La_2O_3 showed little CO oxidation activity.

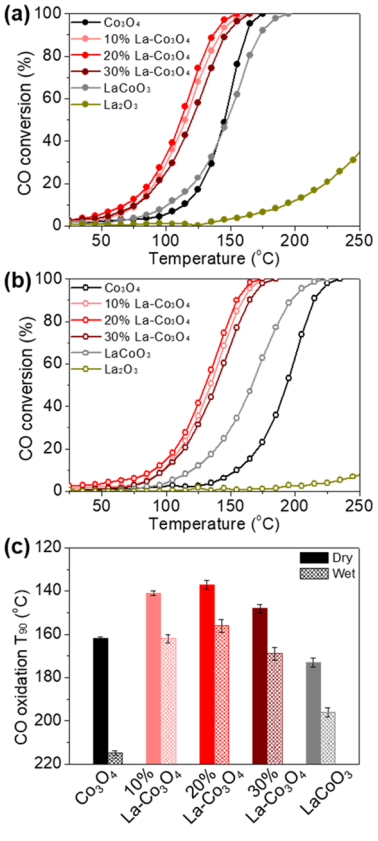


Figure 1. CO oxidation results of La-doped Co_3O_4 catalysts under (a) dry and (b) wet conditions. (c) The temperature at which the CO conversion reached 90% of the catalysts. The reaction under dry condition was conducted at 120 000 mL g^{-1} h^{-1} with a feed stream consisting of 0.4% CO, 10% O_2, and balance He. Under wet condition, the feed gas was passed through a water bubbler at room temperature (~3% of water vapor).

Under wet condition containing ~3% of water vapor, bare Co_3O_4 achieved 90% CO conversion at 215 °C in Figure 1b. The activity was largely degraded compared to the dry condition. However, the La-doped Co_3O_4 catalysts maintained much better activity than bare Co_3O_4. $LaCoO_3$ perovskite also showed better activity than bare Co_3O_4, whereas bare La_2O_3 showed significant degradation in activity under moisture-rich condition. Figure 1c shows T_{90}, which is the temperature at which the CO conversion reached 90%, with error bars. T_{20} and T_{50}, which are the temperatures at which the CO conversion reached 20 and 50%, respectively, are also shown in Figure S2, and they had the same trend as T_{90}. While bare Co_3O_4 showed ΔT_{90} of 53 °C, 20% La-doped Co_3O_4 showed much smaller ΔT_{90} of 19 °C. Figure S3 shows a comparison of 20% La-doped Co_3O_4, bare Co_3O_4, and a physical mixture of Co_3O_4 and La_2O_3. While the physical mixture (20 mol % La) showed relatively improved water resistance (ΔT_{90} = 33 °C), probably due to the hydrophobic nature of La_2O_3,[28,42] the La-doped Co_3O_4 exhibited the most improved activity and water resistance. The water content in the reactant gas was varied to 0, 1, 3, and 10%, as shown in Figure S4. In both La-doped Co_3O_4 and bare Co_3O_4, the activity was degraded in the presence of water, but the difference in activity for the different water contents was rather insignificant. A small amount of water would sufficiently poison the catalyst surface.[22]

Arrhenius plots of 20% La-doped Co_3O_4 and bare Co_3O_4 were obtained for the regions below 20% CO conversion, in which surface reaction determines the overall rate without mass transfer limitation or diffusion limitation, as shown in Figure S5. The apparent activation energies (E_a) of the La-doped Co_3O_4 and bare Co_3O_4 catalysts were 32.4 and 53.3 kJ mol^{-1}, respectively, under dry condition, and the values under wet condition were 40.7 and 70.0 kJ mol^{-1}, respectively. The lower E_a values on the La-doped Co_3O_4 indicate that CO oxidation reactions occur more easily. CO oxidation at the space velocity of 240 000 mL g^{-1} h^{-1} was also performed to see whether the reaction is in the region of mass transfer limitation, as shown in Figure S6. The reaction rate was the same for two different space velocities of 120 000 and 240 000 mL g^{-1} h^{-1}, confirming that this reaction is not in the mass transfer limitation region below 20% CO conversion. Additionally, the reaction rate and the specific rate were estimated at 90 °C, as shown in Figure S7. The specific rate was much higher on the La-doped Co_3O_4 than the bare Co_3O_4 under both dry and wet conditions.

The crystalline structure of the La-doped Co_3O_4 catalysts was examined by XRD, as shown in Figure S8, and the crystallite sizes of the catalysts are listed in Table 1. The diffraction peaks of bare Co_3O_4 showed the spinel structure of Co_3O_4,[14,43] and high-resolution TEM (HR-TEM) images in Figure S9 showed the morphology of the Co_3O_4 nanoparticles.[41,44] All La-doped Co_3O_4 catalysts showed the XRD peaks of both Co_3O_4 spinel structure and $LaCoO_3$ perovskite structure, but La_2O_3 phase was not observed. The Co_3O_4 peaks in the La-doped Co_3O_4 catalysts became smaller and broader as La content increased, indicating that the doped La suppresses the crystal growth of Co_3O_4. The Co_3O_4 phase

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Table 1. Textural Properties of Bare Co_3O_4, La-Doped Co_3O_4, $LaCoO_3$, and La_2O_3 Catalysts

catalyst	surface area ($m^2\ g^{-1}$)	d_{Co} (nm)[a]	surface Co (atom %)[b]	surface La (atom %)[b]	Co^{3+}/Co^{2+}[c]	O_w/O_{latt}[d]	O_{ads}/O_{latt}[d]
Co_3O_4	17	36.4	11.0	0.0	1.14	0.25	0.44
10% La-Co_3O_4	25	15.2	13.6	6.7	1.35	0.04	0.88
20% La-Co_3O_4	28	15.0	14.1	8.5	1.46	0.03	0.99
30% La-Co_3O_4	29	14.7	11.1	9.1	1.43	0.07	1.04
$LaCoO_3$	39	18.6	6.1	12.1	1.40	0.09	1.55
La_2O_3	44		0.0	14.9		0.12	3.89

[a]The crystallite size was calculated by the Scherrer equation from Co_3O_4(311) peak. [b]The atomic percentages of surface Co and La species were obtained from XPS survey data. [c]The area ratio was estimated from the deconvolution of Co 2p peak. [d]The area ratio was estimated from the deconvolution of O 1s peak.

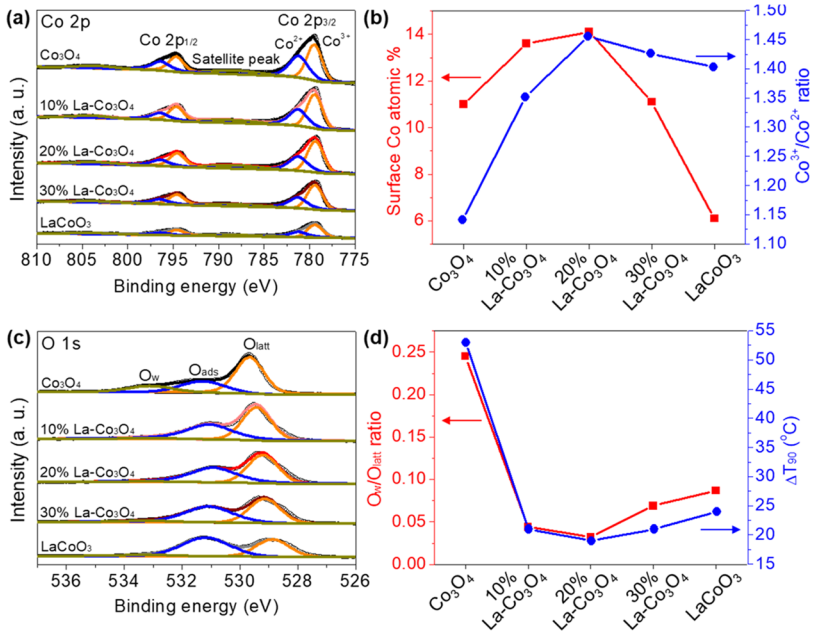


Figure 2. (a) XPS images of Co 2p and (b) the Co atom % and Co^{3+}/Co^{2+} atomic ratio on the surface of La-doped Co_3O_4 catalysts as a function of the La contents. (c) XPS images of O 1s and (d) the O_w/O_{latt} ratio and ΔT_{90} as a function of the La contents.

was transformed gradually into the $LaCoO_3$ phase. The HR-TEM and EDS elemental mapping images in Figure S10 showed that the La species were uniformly distributed on the Co_3O_4 phase.

The chemical states on the surface of La-doped Co_3O_4 catalysts were investigated by XPS in Figure 2, and their quantitative values are shown in Table 1. As shown in Figure 2a, the Co $2p_{3/2}$ peaks of the catalysts could be deconvoluted into Co^{3+} (~779.5 eV), Co^{2+} (~781.2 eV), and their satellite peaks.[11,14,43] The surface Co atom % and Co^{3+}/Co^{2+} ratios on the catalyst surfaces were estimated from the XPS data, then they were plotted as a function of the La contents, as shown in Figure 2b. The surface Co atom % and Co^{3+}/Co^{2+} ratios had a volcano form as La content increased. While bare Co_3O_4 showed a Co^{3+}/Co^{2+} ratio of 1.14, the La-doped Co_3O_4 catalysts showed higher Co^{3+}/Co^{2+} ratios, suggesting that the La cations substitute the Co^{2+} in tetrahedral sites. $LaCoO_3$ showed a Co^{3+}/Co^{2+} ratio higher than bare Co_3O_4, but the surface Co atom % was the lowest, which matched well with the reported values.[45] The 20% La-doped Co_3O_4 had the highest surface Co atom % and Co^{3+}/Co^{2+} ratio, showing that the Co^{3+} cations are present the most on the surface. Interestingly, the trend in the amount of surface Co^{3+} was similar to the trend in activity shown in Figure 1c. The 20% La-doped Co_3O_4 with the most surface Co^{3+} showed the best activity. The rich Co^{3+} active sites on the surface of La-doped Co_3O_4 catalysts resulted in the enhanced activity.

Figure 2c shows XPS O 1s peaks of the catalysts that were deconvoluted into three components at ~529.5, 531.2, and 533.2 eV, assigned to lattice oxygen (O_{latt}), adsorbed oxygen (O_{ads}), and oxygen in molecular water or hydroxyl species adsorbed on the catalyst surface (O_w), respectively.[46−49] The O_w/O_{latt} ratios and ΔT_{90}, which indicate the difference in T_{90} under dry and wet conditions, were plotted as a function of the La contents in Figure 2d. While bare Co_3O_4 showed the O_w/O_{latt} ratio of 0.25, the O_w/O_{latt} ratio was reduced greatly in all La-doped Co_3O_4 catalysts. The formation of surface OH decreased significantly after La doping. Notably, the trend in the amount of surface OH was very similar to the trend in ΔT_{90}, implying that deactivation of the catalysts under wet condition would be caused by water adsorption. The 20% La-doped Co_3O_4 with the smallest amount of surface OH showed the best water resistance.

Figure S11a shows the La 3d XPS data of bare La_2O_3, $LaCoO_3$, and La-doped Co_3O_4 catalysts. The main peaks of the La-doped Co_3O_4 catalysts shifted to higher binding energy, indicating that La interacts strongly with the Co_3O_4 support.[50−52] As shown in Figure S11b, the surface La atom % gradually increases as the La doping content increases. Because the trend in the amount of the surface La atom % does

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not match with the trend in ΔT_{90}, it seems that the surface La atom itself has less effect on the improvement for water resistance.

Rare-earth metals, including La, have been proposed to be inherently hydrophobic in their oxides due to their unique electronic structure.[28] The surface hydrophobicity of various rare-earth metal-doped Co_3O_4 catalysts was investigated by water contact angle measurements.[27] The catalyst powders were uniformly mounted on a carbon tape to form a condensed film, and 10 μL of deionized water was dropped on the film. Varanasi et al. showed that the water contact angles on various REO (La_2O_3, CeO_2, Pr_6O_{11}, Sm_2O_3, Eu_2O_3, Gd_2O_3) were similar at ~105° because the outermost shell is completely filled regardless of the different atomic number, resulting in the similar hydrophobicity.[28] However, Figure S12 shows that the water contact angles on 20% La-, Ce-, Pr-, Sm-, Eu-, and Gd-doped Co_3O_4 were in the range of 97−110°. This difference in the water contact angle strongly indicates that the surface hydrophobicity would not result from the unique electronic structure of REOs. XPS O 1s data in Figure S12 indicate that the surface hydrophobicity rather resulted from different amount of surface OH groups. Figure 3 shows the

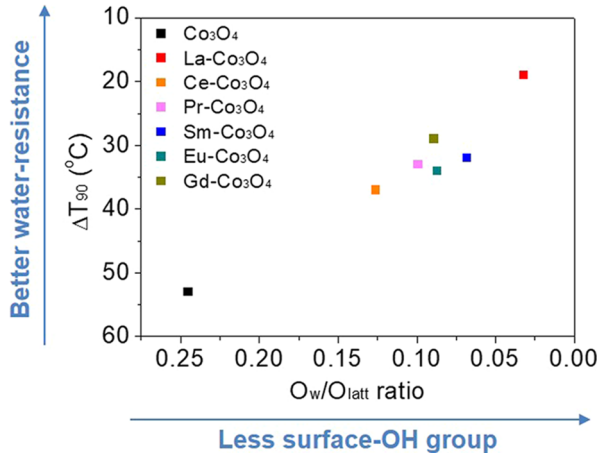


Figure 3. Relation between the O_w/O_{latt} ratio and ΔT_{90} for various rare-earth metal-doped Co_3O_4 catalysts.

relation between the O_w/O_{latt} ratios estimated from XPS O 1s data and ΔT_{90} for various rare-earth metals. A trend was observed showing that ΔT_{90} decreased with better water resistance as the O_w/O_{latt} ratios decreased with less surface OH. Surely, the catalyst with less surface OH under moisture-rich condition presented better water resistance. The full data for CO oxidation on the various rare-earth metal-doped Co_3O_4 under dry or wet conditions are shown in Figure S13.

The formation of surface OH on the catalysts was monitored with in situ DRIFT for CO oxidation under dry and wet conditions, as shown in Figure 4. A large peak at 3435 cm^{-1} is attributed to the stretching vibration of the O−H bonds from surface hydroxyl species.[21,25,45] The peaks at around 2120 cm^{-1} are assigned to gaseous CO and the peaks at 1652 and 1506 cm^{-1} are assigned to bicarbonate species.[21,53] When the catalysts were exposed to water, the surface OH was formed on the bare Co_3O_4, but its formation was greatly reduced on the La-doped Co_3O_4. The peak of bicarbonate species was also reduced on the La-doped Co_3O_4 catalyst. Hydroxyl and bicarbonate species on the Co_3O_4 surface are largely responsible for the deactivation during CO oxidation in the

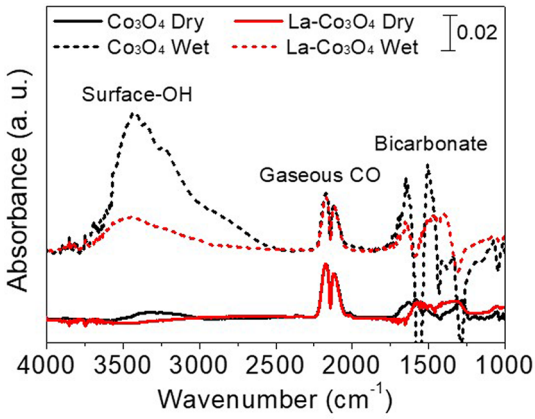


Figure 4. In situ DRIFT spectra of bare Co_3O_4 and 20% La-doped Co_3O_4 catalysts. Under dry condition, the catalysts were exposed to a feed stream consisting of 1% CO, 10% O_2 balanced with Ar for 10 min at room temperature, and then the IR spectrum was obtained. Under wet condition, the feed gas was passed through a water bubbler at room temperature (~3% of water vapor).

presence of water.[12,18,21,22] DRIFT spectra were further obtained after saturating the catalysts with water vapor, as shown in Figure S14. While bare Co_3O_4 showed a large surface OH peak at ~3400 cm^{-1}, the La-doped Co_3O_4 catalysts showed a much smaller peak in the same range. The surface OH peak area and the O_w/O_{latt} ratios were plotted as a function of the La contents in Figure S14b. The less formation of surface OH in the La-doped Co_3O_4 catalysts is consistent with XPS O 1s results. Clearly, the La doping significantly reduces the formation of surface OH on the Co_3O_4 surface, resulting in improved water resistance under moisture-rich condition.

The catalytic activity was further investigated for simultaneous oxidation of CO, C_3H_6, and C_3H_8 under simulated diesel exhaust conditions with 5% of water vapor. The light-off results of the 20% La-doped Co_3O_4, bare Co_3O_4, and commercial DOC are shown in Figure 5a. The commercial DOC contained 1.1 wt % Pt, 0.9 wt % Pd, and 0.1 wt % Rh. The 20% La-doped Co_3O_4 catalyst showed comparable activity to commercial DOC in CO oxidation, while giving lower performance for hydrocarbon oxidation. Long-term stability was also tested by running CO oxidation under dry and wet conditions at 150 °C, as shown in Figure 5b. When water was introduced to the feed, the conversion suddenly dropped from 58.7 to 7.0% in bare Co_3O_4. But the drop in the conversion was greatly alleviated from 99.3 to 82.4% in 20% La-doped Co_3O_4. After 60 h of reaction, only a little degradation was observed; the CO conversion was 51.7% under dry condition and 3.3% under wet condition in bare Co_3O_4, and 97.0% under dry condition and 68.3% under wet condition in 20% La-doped Co_3O_4. Even after the long-term reaction, the La-doped Co_3O_4 catalyst exhibited improved activity and durability.

The effect of La doping on the water resistance of the Co_3O_4 surface was investigated using density functional theory (DFT) calculations. The most stable La doping site was determined on the Co_3O_4(311) surface, which is a dominant surface as shown in the XRD patterns. The location of doped La was estimated for all of the possible positions in the surface unit cell (Figure S1). The doped La is energetically the most likely to occupy the surface Co^{2+} in the tetrahedral site. This is consistent with a higher Co^{3+}/Co^{2+} ratio of the La-doped

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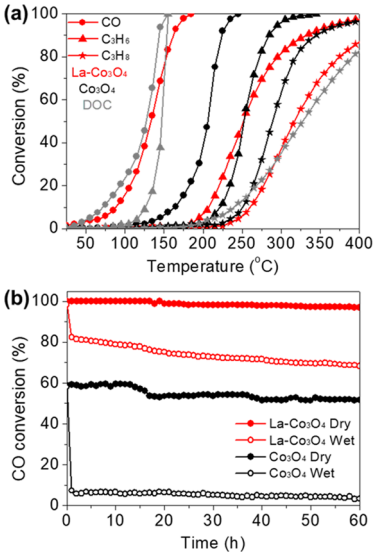


Figure 5. (a) Simulated exhaust oxidation of the 20% La-doped Co_3O_4, bare Co_3O_4, and commercial DOC catalysts. The reaction was conducted at 120 000 mL g^{-1} h^{-1} with a feed flow consisting of 0.4% CO, 0.1% C_3H_6, 0.1% C_3H_8, 5% H_2O, 10% O_2, and balance He. (b) Durability test results over long-term CO oxidation using bare Co_3O_4 and 20% La-doped Co_3O_4 catalysts at 150 °C with and without ~3% of water vapor in the feed flow of 0.4% CO, 10% O_2, and balance He.

Co_3O_4 surfaces as observed in XPS Co 2p results. Wang et al. showed by theoretical study that the surface OH group, which is formed through the dissociative adsorption of H_2O on oxygen vacancy, is much less active than lattice oxygen, resulting in the deactivation of Co_3O_4 for CO oxidation.[22] To get a deeper insight into the less formation of surface OH on the La-doped Co_3O_4 surface, it was first investigated how easily an oxygen vacancy is created on Co_3O_4(311) surface because the vacancy site was reported to facilitate the adsorption of H_2O in a dissociative manner.[22,54−56] Oxygen vacancy formation energy (E_{vf}) was calculated for both bare and La-doped Co_3O_4(311) surfaces, using the most stable configuration of an oxygen vacancy on the surfaces. As described in the left-hand side of Figure 6, the E_{vf} increased by 0.3 eV upon La doping. This indicates that the oxygen vacancy formation is suppressed by La doping.

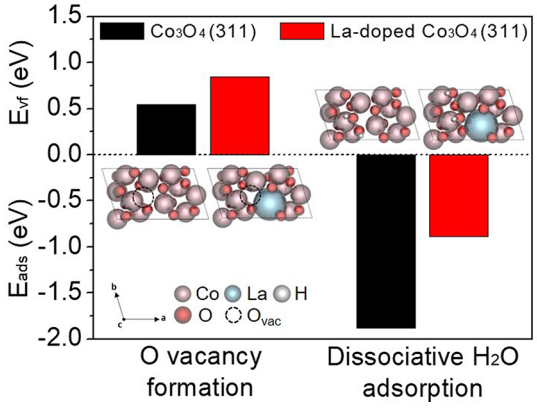


Figure 6. Oxygen vacancy formation energy (E_{vf}) and H_2O dissociative adsorption energy (E_{ads}) of Co_3O_4(311) and La-doped Co_3O_4(311). The optimized surface structure with oxygen vacancy and that with the adsorbed H_2O are shown along with E_{vf} and E_{ads}.

It was further estimated how strongly a H_2O molecule would adsorb on the vacancy site to form two surface OH groups by calculating the dissociative adsorption energy of H_2O (E_{ads}). Each E_{ads} of bare and La-doped surface was calculated on the basis of the most stable configuration of dissociative H_2O adsorption on the oxygen vacancy. As shown in the right-hand side of Figure 6, the energy gain upon the adsorption decreased when La was doped, from −1.88 to −0.89 eV, implying that the surface OH is less likely to be created when La was doped. Our DFT results found that the La doping suppressed the vacancy-assisted dissociative adsorption of H_2O, resulting in less formation of surface OH and thereby mitigating the water poisoning on the Co_3O_4 surface. Thus, the higher ratio of active Co^{3+} and less formation of surface OH on the La-doped Co_3O_4 surface significantly enhanced the catalytic activity under moisture-rich condition.

4. CONCLUSIONS

Doping La into Co_3O_4 surfaces enhanced significantly the activity for CO oxidation under moisture-rich condition. While bare Co_3O_4 was largely deactivated in the presence of water, the La-doped Co_3O_4 catalysts maintained the higher activity better. The increase in surface Co^{3+} on the La-doped Co_3O_4 surface resulted in the enhanced activity. Particularly, the La doping greatly reduced the formation of surface OH on the Co_3O_4 surface, which caused poor water resistance, resulting in enhanced activity under moisture-rich condition. The La-doped Co_3O_4 catalysts presented the activity in CO oxidation as good as commercial DOC with precious metals for simulated exhaust oxidation containing 5% water. DFT calculations revealed that the presence of La dopants made the surface have less oxygen vacancy, suppressing the vacancy-assisted dissociative adsorption of H_2O and the formation of surface OH. This work can provide a novel insight into the surface restructuring for highly active and water-resistant Co_3O_4-based catalysts.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.9b02920.

Optimized structures of the surface unit cells of Co_3O_4(311) and La-doped Co_3O_4(311) (Figure S1); temperature at which the CO conversion reached 20 and 50% of the catalysts (Figure S2); CO oxidation results (Figures S3, S4, S5, S6, S7, and S13); powder XRD patterns (Figure S8); HR-TEM and HAADF-STEM images (Figures S9 and S10); XPS data (Figure S11); relation between the O_w/O_{latt} ratio and water contact angle (Figure S12); DRIFT spectra (Figure S14) (PDF)

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Notes

The authors declare no competing financial interest.

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